

Pragya Singhal

I build the reliable plumbing that agents actually run on.

Software engineer, ~4 years shipping production systems — now building agentic AI: stateful LangGraph agents, grounded RAG, real evaluation harnesses, and the resilient enterprise integration underneath. I ship things that are measured, not demoed.

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0.0 Hallucination rate RAG, 102 served queries	100% Agent tool-selection 50-query eval, guardrails on	0.87 Retrieval hit@k n=30, p50 ~2.0s	0 Failed payments enterprise integration, 30+ users
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// selected work

Agentic AI

side project ·
deployed

CRM Agent

LangGraph · FastAPI · SQLite/WAL · Gradio · HF Spaces

A stateful, tool-calling CRM agent: 6 tools with structured outputs, a prompt-independent human-in-the-loop approval gate for every sensitive action, error recovery (retry-with-backoff + circuit breaker), and cost tracking. Risky actions are safe by design — the gate lives in code, not the prompt.

HITL gate coverage 9/14 → 13/14

tool-selection & params → 100%

50-query eval, 3 runs, fully stable

[live](#) [code](#)

Grounded RAG

side project ·
deployed

HomeoQuiz

Python · Gemini 2.5 Flash-Lite · ChromaDB · sentence-transformers
· HF Spaces

Strict-grounding question-answering over a fixed corpus where every answer cites a source passage, with an automated eval harness (Wilson 95% CIs). A guardrail refuses when the source doesn't cover a topic. Hybrid local-embed + remote-generate — a deliberate cost/latency/rate-limit trade-off.

hallucination 0.0 · quote-validity 1.0

refusals 27/27 correct · over-refuse 2.9%

hit@k 0.87 · p50 ~2.0s

[live](#) [code](#)

LLM in production

production

MyHive Assistant

OpenAI GPT · Firebase · Ionic · Angular · TypeScript · Node.js

A production conversational assistant on a cohort-based wellness marketplace — answers users' health queries and recommends available practitioners. Built personalized exercise recommendations that map each user's mood, pain, sleep and daily cycles to programs in the platform's catalog.

live with real users

team's first user-facing LLM feature

[product](#)

// the plumbing underneath

Before GenAI, four years of backend and integration work — the resilience patterns that agentic systems live or die on.

integration Stripe & Secuconnect via REST +
webhooks — auth, data

migrations Led Codelgniter 3→4 and
DataTables→Tabulator migrations

mapping, idempotent retries, non-happy-path handling. Zero failed payments across 30+ users.

on a large Laravel app with zero regression.

performance Cut page load 2min → 30–40s with custom pagination on a high-traffic booking platform.

team Mentored 5+ junior developers; built and shipped REST APIs across products. First MCP server built.

// stack

GenAI / LLM

RAG

agentic AI

tool calling

structured outputs

workflow orchestration

MCP

evals

guardrails

embeddings

vector DBs

cost/latency opt

frameworks

LangGraph

LangChain

FastAPI

OpenAI

Gemini

Groq

ChromaDB

FAISS

engineering

TypeScript

Python

Node.js

PHP / Laravel

REST API design

MySQL

Docker

HF Spaces

CI/CD